第 5 章練習解答

CH05-E1

1. Cofactors：

$$C_{11} = 1 - 3 = -2,$$
$$C_{12} = -(0 - 6) = 6.$$

第一 row 的第三 entry 為零，所以

$$\det A = 1(-2) + 2(6) = 10.$$

2. 第二 column 加第一 column 的兩倍：

$$B = \begin{bmatrix} 1 & 4 & 0 \\ 0 & 1 & 3 \\ 2 & 5 & 1 \end{bmatrix}.$$

沿第一 row：

$$\det B = 1(1 - 15) - 4(0 - 6) = 10.$$

3. $B = AE$，其中 E 把一個 column 加上另一 column 的倍數，$\det E = 1$。因此

$$\det B = \det A \det E = \det A.$$

幾何上是 shear，body volume ratio 或 area ratio 不變。

CH05-E2

1. 六個 terms：

$$a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32}$$
$$- a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33}.$$

2. Column orders

$$(1, 2, 3), \quad (2, 3, 1), \quad (3, 1, 2)$$

是 even；其餘三個是 odd。

3. 取

$$M = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 3 & 1 \\ 1 & 0 & 4 \end{bmatrix}.$$

六個 terms 中只有

$$2 \cdot 3 \cdot 4 = 24$$

與

$$1 \cdot 1 \cdot 1 = 1$$

非零，均為 even，因此

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第 5 章練習解答 （續）

$$\det M = 25.$$

CH05-E3

1.

$$C_{11} = 10, \qquad C_{12} = 4, \qquad C_{13} = -2.$$

2.

$$\det A = 2(10) + 1(4) + 0(-2) = 24.$$

3. 第二 row cofactors：

$$C_{21} = -(4) = -4,$$
$$C_{22} = 8, \qquad C_{23} = -(4) = -4.$$

所以

$$\det A = (-1)(-4) + 3(8) + 1(-4) = 24.$$

CH05-E4

1. $\det A = 10$。Cofactor matrix 為

$$C = \begin{bmatrix} -2 & 6 & -2 \\ -2 & 1 & 3 \\ 6 & -3 & 1 \end{bmatrix}.$$

$$\operatorname{adj} A = C^T = \begin{bmatrix} -2 & -2 & 6 \\ 6 & 1 & -3 \\ -2 & 3 & 1 \end{bmatrix}.$$

$$A^{-1} = \frac{1}{10} \begin{bmatrix} -2 & -2 & 6 \\ 6 & 1 & -3 \\ -2 & 3 & 1 \end{bmatrix}.$$

2. Cofactor identities 逐 row 給 diagonal 10，off-diagonal 0：

$$A \operatorname{adj} A = 10I.$$

3. 若 $Ax = b$，

$$x = \frac{1}{\det A} \operatorname{adj}(A) b.$$

正確但 costly。$n \times n$ adjugate 有 n^2 個 minors；大矩陣應使用 pivoted factorization。

CH05-E5

1.

$$\det A = -2, \qquad \det A_1 = -4, \qquad \det A_2 = 2.$$

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$$x_1 = 2, \qquad x_2 = -1.$$

2. Verification：

$$3(2) + 4(-1) = 2, \qquad 5(2) + 6(-1) = 4.$$

3. 若 $b = (2, 5)^T$：

$$\det A_1 = -8, \qquad \det A_2 = 5,$$

$$x_1 = 4, \qquad x_2 = -\frac{5}{2}.$$

CH05-E6

1.

$$P = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, \qquad \det P = 5.$$

Vectors 的 counterclockwise order 與 positive orientation 一致，signed area 與 area 都是 5。

2. 令

$$X = w_2 z_3 - w_3 z_2,$$

$$Y = w_3 z_1 - w_1 z_3, \qquad Z = w_1 z_2 - w_2 z_1.$$

直接展開：

$$w_1 X + w_2 Y + w_3 Z = 0,$$

每一項均成對抵消。同理 $z \cdot (w \times z) = 0$。

3.

$$w \times z = (-2, 1, 1)^T.$$

Triple product identity：

$$\det[w \; z \; w \times z] = \|w \times z\|^2 = 6.$$

CH05-E7

1. cA 的 n 條 columns 各提出 c：

$$\det(cA) = c^n \det A.$$

2.

$$1 = \det(Q^T Q) = (\det Q)^2,$$

所以

$$\det Q = \pm 1, \qquad |\det Q| = 1.$$

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3. $S^T = -S$，若 n odd：

$$\det S = \det(S^T) = \det(-S) = (-1)^n \det S = -\det S.$$

因此 $\det S = 0$，奇數維 skew-symmetric matrix singular。

n even 時 $(-1)^n = 1$，推論不成立。反例：

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

determinant 是 1。

CH05-E8

1. 沿第一 row 展開：

$$M_{11} = \begin{bmatrix} d & 1 \\ 2 & a+d \end{bmatrix},$$

$$\det M_{11} = d(a+d) - 2.$$

$$M_{12} = \begin{bmatrix} c & 1 \\ 0 & a+d \end{bmatrix},$$

$$\det M_{12} = c(a+d).$$

第三 entry 為 zero，所以

$$\det A = a[d(a+d) - 2] - bc(a+d).$$

2. 取 $a = d = 0, b = c = 1$：

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}.$$

第一與第三 columns 都是 $(0,1,0)^T$，rank < 3，det 為 zero。幾何上 three columns 在二維 subspace，體積為 zero。

3. $a = d = 1, b = c = 0$：

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 2 & 2 \end{bmatrix}.$$

$$\det A = 1(2-2) = 0.$$

Trace 是 $1+1+2 = 4$。Eigenvalues 的乘積是 determinant 0，所以至少一個 eigenvalue 為 0；rank 不足與此一致。

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第 5 章練習解答 （續）

原創驗算程式附錄: determinants.py

```
"""Recompute the numerical checks in the chapter five notes."""

from itertools import permutations
from math import isclose


Matrix = tuple[tuple[float, ...], ...]


def determinant(matrix: Matrix) -> float:
    n = len(matrix)
    if n == 0:
        return 1.0
    total = 0.0
    for permutation in permutations(range(n)):
        inversions = sum(
            permutation[i] > permutation[j]
            for i in range(n)
            for j in range(i + 1, n)
        )
        product = 1.0
        for row, column in enumerate(permutation):
            product *= matrix[row][column]
        total += (-1 if inversions % 2 else 1) * product
    return total
```

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```
def minor(matrix: Matrix, row: int, column: int) -> Matrix:
    return tuple(
        tuple(value for j, value in enumerate(values) if j != column)
        for i, values in enumerate(matrix)
        if i != row
    )


def cofactors(matrix: Matrix) -> Matrix:
    return tuple(
        tuple(
            (-1) ** (i + j) * determinant(minor(matrix, i, j))
            for j in range(len(matrix))
        )
        for i in range(len(matrix))
    )


def transpose(matrix: Matrix) -> Matrix:
    return tuple(zip(*matrix))


def multiply(left: Matrix, right: Matrix) -> Matrix:
    return tuple(
        tuple(
```

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```
            sum(left[i][k] * right[k][j] for k in range(len(right)))
            for j in range(len(right[0]))
        )
        for i in range(len(left))
    )


def cramer(matrix: Matrix, right: tuple[float, ...]) -> tuple[float, ...]:
    base = determinant(matrix)
    if isclose(base, 0.0, abs_tol=1e-12):
        raise ValueError("Cramer's rule needs a nonsingular matrix")
    answers = []
    for column in range(len(matrix)):
        changed = [list(row) for row in matrix]
        for row in range(len(matrix)):
            changed[row][column] = right[row]
        answers.append(determinant(tuple(tuple(row) for row in changed)) / base)
    return tuple(answers)


a = (
    (1.0, 2.0, 0.0),
    (0.0, 1.0, 3.0),
    (2.0, 1.0, 1.0),
)
assert isclose(determinant(a), 10.0)

cofactor_matrix = cofactors(a)
adjugate = transpose(cofactor_matrix)
assert multiply(a, adjugate) == tuple(
    tuple(10.0 if i == j else 0.0 for j in range(3))
    for i in range(3)
)

b = (1.0, 2.0, 3.0)
solution = cramer(
    ((3.0, 4.0), (5.0, 6.0)),
    (2.0, 4.0),
)
assert solution == (2.0, -1.0)

u = (2.0, 1.0)
v = (1.0, 3.0)
signed_area = u[0] * v[1] - u[1] * v[0]
assert signed_area == 5.0

print("det(A) =", determinant(a))
print("A adj(A) =", multiply(a, adjugate))
print("Cramer solution =", solution)
print("signed area =", signed_area)
```

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參考資料與署名

Gilbert Strang. Introduction to Linear Algebra, 5th Edition, 2016.

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